

# **Retail Profit Leakage & Growth Optimization System**

## **Final Master Project Report – End-to-End Data Analytics Project**

### **1. Project Introduction**

This project focuses on identifying profit leakage in a retail business using data analytics techniques.

The objective of the project is to analyze transactional retail data to understand the factors that influence profitability.

The analysis combines multiple tools including Python, SQL, Excel, and Power BI to create a complete analytics pipeline.

The final outcome of the project is a set of actionable insights and an executive dashboard that helps decision makers optimize business performance.

## 2. Dataset Description

The dataset used in this project is the Superstore retail dataset which contains detailed order level transactional data.

It includes sales, profit, discount, quantity, customer information, product categories, and regional information.

The dataset contains 9,994 transactions and spans the period between 2014 and 2017.

This dataset is commonly used for business intelligence and analytics learning because it represents realistic retail operations.

| Attribute     | Details                           |
|---------------|-----------------------------------|
| Dataset Name  | Superstore Retail Dataset         |
| Total Rows    | 9,994 Transactions                |
| Total Columns | 21 Columns                        |
| Time Period   | 2014 – 2017                       |
| Business Type | Retail Sales                      |
| Key Metrics   | Sales, Profit, Discount, Quantity |

### **3. Business Problem**

Retail businesses often face profit leakage due to excessive discounting, inefficient pricing strategies, and loss making product categories.

Understanding how different variables affect profitability is essential for making data driven decisions.

This project aims to identify patterns that lead to loss making transactions and recommend strategies to improve overall profitability.

The analysis focuses on category performance, regional performance, customer contribution, and discount impact.

## **4. Day 1 – Data Cleaning and Preparation (Python)**

The first phase of the project focused on preparing the dataset for analysis using Python.

The dataset was imported using Pandas and explored to understand its structure and column types.

Data quality validation included checking for missing values and duplicate records.

The Order Date and Ship Date columns were converted to datetime format to enable time based analysis.

Feature engineering was performed to create additional analytical columns such as Profit Margin, Loss Flag, Year, and Month.

### **Key Tasks Performed in Day 1**

Importing Python libraries such as Pandas and NumPy.

Loading the dataset using `read_csv()`.

Checking dataset structure using `head()`, `info()`, and `describe()`.

Validating data quality using `isnull()` and `duplicated()`.

Creating new analytical features to support business analysis.

## 5. Day 2 – Exploratory Data Analysis (Python)

Exploratory Data Analysis was performed using Python visualization libraries including Matplotlib and Seaborn.

The goal of EDA was to identify patterns, trends, and anomalies in the dataset.

Multiple charts were created to analyze category performance, regional performance, and discount impact.

The analysis helped identify which categories generate the highest profit and which products generate losses.

### Key EDA Visualizations

Sales by Category

Profit by Category

Profit by Sub Category

Monthly Sales Trend

Profit by Region

Discount vs Profit Scatter Plot

Correlation Heatmap

Profit Distribution Histogram

Sales vs Profit Scatter Plot

Discount Distribution Box Plot

## 6. Day 3 – SQL Business Analysis

The cleaned dataset was imported into PostgreSQL to perform business analysis using SQL queries.

SQL queries were used to aggregate data and identify profitability patterns.

Approximately fifteen analytical queries were written including customer analysis, regional performance, and product profitability.

Window functions such as RANK() were used to rank regions based on profit.

### SQL Techniques Used

Aggregation functions such as SUM(), AVG(), and COUNT().

Grouping operations using GROUP BY.

Filtering using WHERE conditions.

Sorting using ORDER BY.

Advanced analytical queries using window functions.

## **7. Day 4 – Excel Business Analysis**

Microsoft Excel was used to perform pivot based analysis and scenario modelling.

Pivot tables were created to analyze sales performance by region, category, and segment.

Conditional formatting was used to highlight loss making transactions.

Charts were used to visualize trends and patterns.

### **Excel Analysis Performed**

Sales by Region

Profit by Category

Sales by Segment

Sub Category Profit Analysis

Monthly Sales Trend

Top Customers by Sales

Discount vs Profit Analysis

Scenario Modelling for Pricing Strategy

## **8. Day 5 – Power BI Dashboard**

A Power BI dashboard was developed to present business insights in an interactive format.

DAX measures were created to calculate Total Sales, Total Profit, Profit Margin, and Total Orders.

The dashboard includes multiple pages including Executive Overview, Profit Leakage Analysis, Customer Analysis, and Regional Performance.

Interactive slicers allow users to filter the dashboard by region, category, segment, and year.

### **Power BI Visualizations**

KPI Cards

Bar Charts

Line Charts

Scatter Charts

Pie Charts

Map Visualizations

Tables for loss making products



## 9. Key Business Insights

Technology category generates the highest overall revenue and profit.

Western region produces the highest profit among all regions.

Tables and Bookcases are consistently loss making sub categories.

Consumer segment contributes the largest share of revenue.

High discount levels strongly correlate with reduced profitability.

## **10. Business Recommendations**

Reduce excessive discounts on low margin products.

Review pricing strategy for loss making product groups.

Focus marketing investment on profitable regions.

Monitor performance continuously using BI dashboards.

Adopt data driven decision making processes.

## **11. Skills Demonstrated**

Python Data Cleaning and Feature Engineering

Exploratory Data Analysis using Python

SQL based business analysis

Excel pivot analysis and scenario modelling

Power BI dashboard development

Business intelligence storytelling